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[Paper 4] Paper "Disk" (published) and "Disk-II" - Isolated Disk Run

This page presents results from the AGORA isolated disk simulations being led by Agertz, Teyssier, Wadsley, etc. (formerly AGORA Paper "4").

If you want to read about the pilot studies for this comparison, please visit the archived [OLD PAGE](#).

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The Isolated disk simulation data by 9 participating code groups have been made public as of Jan. 2020. Please find them in the [Quick Link](#) page.

If you visit this page while participating in the Working Sessions of the [5th AGORA Workshop](#), please directly go to [Section 6](#)!

[1. Prerequisites for Paper "4" Participants](#)

- (1) AGORA isolated disk ICs: [Dropbox link](#) with [README.txt](#), also described in Section 2.2 of [Flagship paper](#)
- (2) Introduction to AGORA isolated disk comparison: [Talk](#) by Oscar Agertz in the 3rd Workshop, Aug. 2014
- (3) Up-to-date participants list: See the table in the most recent progress report ([here](#) as of Aug. 2016)

[2. Future Plans \(Summary from 3rd Workshop+7th Web Conference\)](#)

[\(1\) Milestones](#)

- (1) Have all codes use Grackle common physics package (see Paper "3"; one-zone tests desired)
- (2) Run the low resolution disk setups
- (3) Have the yt pipeline in place for most of the analysis (see the shopping list)
- (4) Transfer of snapshots to NERSC as they become available
- (5) Working group telecon to be arranged, share plots and datasets via Workspace and on NERSC
- (6) Push out the Paper

(2) Dataset Requirements

(2-1) Requirements for "First" Datasets (Grackle+noSF; Deadline May 31) (red color: added on Aug. 23, 2015)

- We agreed on the specific requirements for the "first" datasets to be submitted. All these and more discussed and documented in the Dropbox [README file](#).

(a) No SF

(b) Low-resolution (i.e. $M_{\text{gas}}=8.593 \times 10^4 M_{\text{sun}}$ or $dx=80$ pc, described in the readme file; we discussed how the gravitational softening and hydro smoothing kernel length in modern simulations are different by ~factor of 10; people agreed to try cases using both synchronized and mismatched values to understand the difference and how it affects the density PDF)

(c) Grackle cooling (tabulated primordial cooling by Cloudy at Z_{sun} + tabulated metal cooling by Cloudy + UV heating by Haardt & Madau 2012; see [Grackle doc](#) for more info)

(d) Jeans pressure floor (described in the readme file **but with pressure support with $\Delta x = 80$ pc for SPH codes**; this is to avoid artificial Jeans fragmentation due to unresolved pressure gradients; this may not be needed after all, probably due to the non-thermal support from turbulence especially if feedback is included)

(e) Consensus minimum smoothing length for SPH ($= 0.2 \cdot \text{eps}_{\text{grav}}$, where $\text{eps}_{\text{grav}} = 80$ pc is the gravitational softening length = equivalent Plummer softening length)

(f) Consensus refinement criteria for AMR (refine once a gas cell contains 1–8x the SPH gas particle mass or 8 collisionless -- DM or IC stars in disk/bulge -- particles) + corrected circular velocity file in the IC (vcirc_SPH.dat for AMR codes; see the readme file for reasoning)

(g) Initial metallicity of 0.02041 ($= 1 Z_{\text{sun}}$ in Grackle v2.0, but $= 1.5761 Z_{\text{sun}}$ in Grackle v2.1)

- One-zone/one-particle test for Grackle cooling? – Various benchmarks exist in [Grackle 2.1](#) (as of June 2, 2015; test outputs a t–T plot), in page 14/15 of [Oscar's talk](#), and in Nick's Grackle API.
- Possible issues in Grackle with (1) mean molecular weight and (2) solar metallicity? – Resolved in [Grackle 2.1](#). See detailed discussion [here](#).
- Submit "first" datasets for preliminary comparison and let Ji-hoon know.
 - For starters, submit only two snapshots at 0 Gyr and at 0.5 Gyr. But make sure to store at every 50 Myrs for future analysis. If you don't have a user account at NERSC check [here](#).
 - Ji-hoon, Cameron, and other volunteers can run some initial cross-platform analysis, and already work with yt representatives to resolve outstanding issues. The [agora-analysis-script](#) repository is being updated as the analysis progresses. Start with the two basic diagnostics: (A) gas surface density map + (B) rho–T phase plot.
 - Produce multi-panel images and plots on the web, and iterate until each code group arrives at a parameter configuration that makes its code compatible with others.
- Resolution comparison will be carried out by a subset of groups: find an optimal combination of refinement parameters to make AMR compatible with SPH (Ben Keller & Robert Feldmann volunteered).
- We will eventually use median-resolution runs ($8.593 \times 10^4 M_{\text{sun}}$ or 40 pc) for the Paper. Stellar feedback schemes to be used in cosmological runs will be calibrated at this resolution. --> This is still to be discussed as we move the Paper forward. May be postponed for future Paper (note added in Jun. 2016).
- Check out [this Google Doc \(Paper "4" Working Session Meeting Notes\)](#) to see the full context of discussion during the 4th AGORA Workshop.

(2-2) Requirements for "Second" Datasets (Grackle+SF+ThermalFbck; Deadline June 30) (red color: added on Aug. 23, 2015)

- We agreed on the specific requirements for the "second" datasets to be submitted. All these and more discussed and documented in the Dropbox [README file](#).

(a) Simple Schmidt-like SF recipe, $d(\rho_*)/dt = e_{\text{ff}} \cdot \rho_{\text{gas}} / t_{\text{ff}}$, described in the readme file and the Flagship Paper: With a density threshold $n_{\text{SF}}=10$ H/cc for the low resolution case, and with $e_{\text{ff}}=1\%$ and free-fall time $t_{\text{ff}} = \sqrt{3\pi/(32 \cdot G \cdot \rho_{\text{gas}})}$. **Minimum star particle mass = gas particle mass in SPH.**

(b) Thermal feedback of Type II SN only with $1e51$ ergs/SN: **1 SN explosion event per star particle (i.e. no stochastic explosion). Per 91 Ms of stellar mass 1 SNII occurs, and 2.63 Ms of metals and 14.8 Ms of gas (including metals) are released. Instantaneous injection after 5 Myr lifetime. Deposit schemes left at participants' discretion.**

- Check if SF+ThermalFbck recipes in each codes are compatible within SPH/AMR peers. Submit "second" datasets as they become available. Another round of iteration process with each groups.

- Need to constrain the Grackle version to be used (v2.1 versus v2.0)? **Answer: No need to constrain this. Keep using what you have used.**

- Need to include subgrid multiphase ISM model for the gas above SF threshold? (e.g. Springel & Hernquist 2003; raised by Nagamine, Lupi) **Answer: No.**
- Check out [this Google Doc \(Paper "4" Working Session Meeting Notes\)](#) to see the full context of discussion during the 4th AGORA Workshop.

(2-3) Requirements for "Third" Datasets (Grackle+SF+FavoriteFbck; Deadline TBD)

- We will tentatively focus on comparisons of 2 types of runs at the moment: Grackle+no SF run, and Grackle+SF+thermal fbck run. But we may consider Grackle+SF+FavoriteFbck runs as a subset comparison. This will become an important component for understanding the full AGORA runs.
- We will calibrate our feedback model so it has an impact at galactic scale.
- Figure-of-merit to decide which galaxy looks reasonable?

(3) Analysis To Carry Out

The following list of analysis was agreed to be of interest, but we will reconsider the list as we move along. Colored in red are the ones we plan to focus in the summer of 2016, as a result of our discussion in our [web conference](#) on Jun. 29, 2016.

- Density maps of stars and gas
 - Disk structure: gas and SFR surface density, profiles, vertical structure of each components, scale radius/height
 - Density and temperature PDF (crucial to understand phase structure), density DF
 - Star formation histories
 - **Radius vs velocity dispersion (gas and stars), V/σ evolution** (Cameron)
 - **Star formation rate surface density**
 - **Kennicutt-Schmidt relation**
 - **Clumps detection and statistics (self-gravitating gas and stars, mass spectrum etc.)** (Cameron)
 - Outflows/inflow: mass/metal flux (spherical/cylindrical symmetry)
 - Mass, momentum, entropy and energy conservation/evolution, effect of time-stepping strategy (Cameron)
 - Disk stability analysis (e.g. Q_{gas} , Q_{stars} + multicomponent) (Hui)
 - Bulge-to-disk decomposition (Santi)
- below are new additions since the 2014 Workshop -----
- **Gas disk elevation map (average z along lines of sight on a x - y plane)**
 - **Radius vs gas disk average vertical height ($\sim$ thickness)**
 - **Resolution comparison: grid size for AMR, h_{smh} or $[M/\rho]^{1/3}$ for SPH, time evolution? resolution study with med-resolution runs at least by some codes?**
 - **Fourier modes**
 - Radius vs specific angular momentum
 - Spirals and bar formation
 - Porosity of the ISM (i.e. volume filling fraction of dense, star-forming gas? Hot diffuses gas? How stable/unstable are these quantities over time? How does it change as a function of density? AMR vs SPH?)

3. Pilot Study (Late 2013 - Early 2015)

Before enforcing well defined cooling/heating routines, star formation prescriptions and feedback models, we want to carry out a quick pilot study in order to get the comparison started. We do this to start off the project in a simple way in order to iron out any kinks in the pipelines. Furthermore, this will spark conversation among us and allow us to better understand how we want to guide the proper comparison project at a later stage...

READ MORE IN THE ARCHIVED [OLD PAGE](#).

4. "First" Datasets (Grackle+noSF)

Below we post the first batch of compilation figures we agreed to produce first. This is hardly the final result of the comparison, but only the beginning of our data analysis phase. A real discussion can now begin during and after the [4th AGORA Workshop](#) (Aug. 21–23, 2015) about further analysis, additional comparison, and code differences, now that we have carefully formed a baseline to compare different codes at all.

Please read the [Code Notes](#) below carefully before drawing any conclusion. For example, note the distinction between pressure floor implementations in different codes, which were left at participants' discretion with an approximate guideline. Note also that some figures may change significantly as we identify and fix various issues. Reload the images to make sure you are looking at the most recent ones. We also kindly remind you of the [internal policy](#) regarding the disclosure of AGORA simulation results.

The [agora-analysis-script](#) repository is being updated for the common yt script used to carry out the analysis presented here ([script-Paper4-disk-new.py](#)). Check out the script to learn exactly what quantities are drawn. We recommend to use the most recent yt-dev so that the script works as intended. The recently released stable yt-3.2.0 ([announcement](#) on July 24, 2015, thanks in part to AGORA members including Arraki, Keller, Smith, Turk) may fail to work in some cases. Feel free to suggest improvements on the script, or add functionalities to it.

FYI, see [this Google Doc \(Paper "4" Working Session Meeting Notes\)](#) for the notes of discussions, show-and-tells, comments on figures during the [4th AGORA Workshop](#).

[\(1\) Gas Surface Density Maps](#)

- [0 Myr compilation](#) (first posted: 08/07/2015, last update: 09/23/2016)
- [500 Myr compilation](#) (first posted: 08/07/2015, last update: 09/23/2016)

• Notes: 0 Myr comparison is to check the compatibility of ICs interpreted by each codes. Each frame is centered on the location of maximum gas density within 1 kpc from the center of gas mass. yt parameters `n_ref=8` and `over_refine_factor=2` are used for visualizations for SPH codes (same for temperature, resolution, metallicity maps below); see the [agora-analysis-script](#) for details. As of Jun. 2016, this map is produced temporarily with a stable yt-3.2.3 (*not* yt-3.3-dev), due to unresolved issues likely related to yt [PR #1740](#).

[\(2\) Temperature Projection Maps](#)

- [0 Myr compilation](#) (first posted: 08/07/2015, last update: 09/23/2016)
- [500 Myr compilation](#) (first posted: 08/07/2015, last update: 09/23/2016)

• Notes: Density-square weighted. A hot gas halo exists only in AMR codes (see README.txt above). As of Jun. 2016, this map is produced temporarily with a stable yt-3.2.3 (*not* yt-3.3-dev), due to unresolved issues likely related to yt [PR #1740](#).

[\(3\) Density-Temperature PDF](#)

- [0 Myr compilation](#) (first posted: 08/07/2015, last update: 09/23/2016)
- [500 Myr compilation](#) (first posted: 08/07/2015, last update: 09/23/2016)

• Notes: Raw particle fields are used for SPH codes, not the interpolated or smoothed fields in yt. A thick dashed line in each panel to plot the mean temperature in each density bin for CHANGA. The thin dotted lines denote the slope of constant pressure process, and the thin dot-dashed lines that of constant entropy process.

[\(4\) Radius-Gas Rotational Velocity PDF + Rotation Curves](#)

- [0 Myr compilation](#) (first posted: 08/10/2015, last update: 09/23/2016)
- [500 Myr compilation](#) (first posted: 08/10/2015, last update: 09/23/2016)
- [500 Myr rotation curves](#) (first posted: 08/11/2015, last update: 09/23/2016)

• Notes: Rotational velocity of gas. Radius and velocity with respect to the galactic center (location of maximum gas density within 1 kpc from the center of gas mass for each snapshot). This is to check if we established an access to all phase-space information. Raw particle fields are used for SPH codes, not the interpolated or smoothed fields in yt. The third figure is simply the composite of all the mass-weighted average lines in the second figure. For AMR codes, only dense enough cells are drawn. yt-dev (post yt-3.2.0) needed for the common yt script to properly produce this PDF ([PR #1684](#), and [PR #2097](#) for the yt issue tracker [#1203](#)).

[\(5\) Radially-binned Gas Velocity Dispersion](#)

- [500 Myr velocity dispersion curves](#) (first posted: 07/03/2016, last update: 09/23/2016)
- [500 Myr vertical velocity dispersion curves](#) (first posted: 08/22/2016, last update: 09/23/2016)

• Notes: Velocity dispersion of gas. This quantifies the residual velocity components other than the rotational velocity calculated above; in other words, square roots of mass-weighted averages of $[v_i - v_{\text{rot, local}}]^2$. Radius and velocity with respect to the galactic center (location of maximum gas density within 1 kpc from the center of gas mass for each snapshot). Raw particle fields are used for SPH codes, not the interpolated or smoothed fields in yt. For AMR codes, only dense enough cells are used. The second figure shows only the vertical velocity dispersion to illustrate the contribution in dispersion by the vertical gas movement. yt-dev (post yt-3.2.0) needed for the common yt script to properly produce this PDF ([PR #1684](#), and [PR #2097](#) for the yt issue tracker [#1203](#)).

[\(6\) Radius-Gas Vertical Height PDF + Radially-binned Average Gas Vertical Height](#)

- [0 Myr compilation](#) (first posted: 07/01/2016, last update: 09/23/2016)
- [500 Myr compilation](#) (first posted: 07/01/2016, last update: 09/23/2016)
- [500 Myr average gas height curves](#) (first posted: 07/01/2016, last update: 09/23/2016)

• Notes: Vertical height of gas from the disk plane. Height is the vertical distance -- absolute value, in other words, $|z_i - z_{\text{center}}|$ -- from the disk x-y plane centered on the galactic center (location of maximum gas density within 1 kpc from the center of gas mass for each snapshot). Radius with respect to the galactic center. Raw particle fields are used for SPH codes, not the interpolated or smoothed fields in yt. The third figure is simply the composite of all the mass-weighted average lines in the second figure.

[\(7\) Gas Mass Distribution in Density \(Density PDF\)](#)

- [0 Myr compilation](#) (first posted: 08/11/2015, last update: 09/23/2016)
- [500 Myr compilation](#) (first posted: 08/11/2015, last update: 09/23/2016)

• Notes: Gas mass distributions along the density axis, simply derived from (3) above. Raw particle fields are used for SPH codes, not the interpolated or smoothed fields in yt.

[\(8\) Gas Mass Distribution in Radius + Radially-binned Gas Surface Density](#)

- [0 Myr compilation - mass](#) (first posted: 08/12/2015, last update: 09/23/2016)
- [500 Myr compilation - mass](#) (first posted: 08/12/2015, last update: 09/23/2016)
- [0 Myr compilation - surface density](#) (first posted: 08/12/2015, last update: 09/23/2016)
- [500 Myr compilation - surface density](#) (first posted: 08/12/2015, last update: 09/23/2016)

• Notes: The first two figures are gas mass distributions (accumulated) along the cylindrical radius axis. The last two figures are gas surface density profiles as functions of cylindrical radius, simply derived from the first two. Radius with respect to the galactic center (location of maximum gas density within 1 kpc from the center of gas mass for each snapshot). Raw particle fields are used for SPH codes.

[\(9\) Gas Mass Distribution in Height + Vertically-binned Gas Surface Density](#)

- [0 Myr compilation - mass](#) (first posted: 08/13/2015, last update: 09/23/2016)
- [500 Myr compilation - mass](#) (first posted: 08/13/2015, last update: 09/23/2016)
- [0 Myr compilation - surface density](#) (first posted: 08/13/2015, last update: 09/23/2016)
- [500 Myr compilation - surface density](#) (first posted: 08/13/2015, last update: 09/23/2016)

• Notes: The first two figures are gas mass distributions (accumulated) along the vertical height axis. The last two figures are gas surface density profiles as functions of vertical height, simply derived from the first two. Raw particle fields are used for SPH codes. Height with respect to the galactic center (location of maximum gas density within 1 kpc from the center of gas mass for each snapshot).

[\(10\) Gas Cell Size Maps](#)

- [0 Myr compilation](#) (first posted: 08/26/2015, last update: 09/23/2016)
- [500 Myr compilation](#) (first posted: 08/26/2015, last update: 09/23/2016)

• Notes: Projection of gas cell_size weighted by $(\text{cell_volume})^{-2}$ so that the maximally refined cells along the lines of sight could stand out. For SPH codes, the octtree structure naturally built by yt is used (cells refined by 2 in each axis until a cell has less than 8 particles). The color range spans from 50 to 500 pc in a logarithmic scale. As of Jun. 2016, this map is produced temporarily with a stable yt-3.2.3 (*not* yt-3.3-dev), due to unresolved issues likely related to yt [PR #1740](#).

[\(11\) Gas Resolution Maps](#)

- [0 Myr compilation](#) (first posted: 07/01/2016, last update: 09/23/2016)
- [500 Myr compilation](#) (first posted: 07/01/2016, last update: 09/23/2016)

• Notes: For AMR codes, this is a projection of gas cell_size, so the same figure as (10) above. For SPH codes, this is a projection of gas particle_size -- defined as $(\text{mass}/\text{density})^{1/3}$ -- smoothed on yt's octtree structure (the same smoothing scheme as the surface density projection above). They are weighted by $([\text{cell}/\text{particle}]_{\text{volume}})^{-2}$ so that the maximal resolution elements along the lines of sight could stand out. The color range spans from 50 to 500 pc in a logarithmic scale. As of Jun. 2016, this map is produced temporarily with a stable yt-3.2.3 (*not* yt-3.3-dev), due to unresolved issues likely related to yt [PR #1740](#).

[\(12\) Gas Disk Elevation Maps](#)

- [0 Myr compilation](#) (first posted: 06/30/2016, last update: 09/23/2016)
- [500 Myr compilation](#) (first posted: 06/30/2016, last update: 09/23/2016)

• Notes: Density-weighted averages of gas cells' vertical elevations with respect to the disk x-y plane centered on the galactic center (location of maximum gas density within 1 kpc from the center of gas mass for each snapshot) along the lines of sight. In other words, density-weighted averages of $(z_j - z_{\text{center}})$; thus, a negative value indicates that the cell is located below the disk plane. This is to visualize and estimate the warping/flaring of the gas disk (see e.g. Levine et al. 2006 for an observational study). As of Jun. 2016, this map is produced temporarily with a stable yt-3.2.3 (*not* yt-3.3-dev), due to unresolved issues likely related to yt [PR #1740](#).

[\(13\) More Diagnostics to Come](#)

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[\(14\) Code Notes](#)

CODE	SUBMITTED BY	PRESSURE FLOOR IMPLEMENTATION	OTHER NOTES (Important recent updates colored blue.)
ART-I	Ceverino	• Agertz formula with $N_{\text{Jeans}} = 4$ (note 1).	• Grackle v1.1 (compatible with v2.0) • yt-dev or yt-3.2.0 needed for the common yt script to properly work with ART-I data (PR #1438). • The above PR is essential for retrieving temperature from the internal energy field ART-I tracks. • The ART-I entry was last updated on Sep. 23, 2016.
ART-II	Feldmann	• Actual gas pressure or temperature field is never touched. Changes are made only to the pressure that goes into the Riemann solver, with $P_{\text{cell}} = \max(P_{\text{Jeans}}, P_{\text{gas}})$, only at the maximum refinement level. • $P_{\text{Jeans}} = n_{\text{J}} k_B T_{\text{Jeans}}$, with $T_{\text{Jeans}} = T_{\text{J}} (n_{\text{H}}/n_{\text{J}})$, where $T_{\text{J}} = 1800 \text{ K}$, $n_{\text{J}} = 8 \text{ H/cc}$, and $n_{\text{H}} = \rho/m_{\text{H}}$ • This polytrope is designed to match the Agertz formula with $N_{\text{Jeans}} \sim 4$.	• Grackle v1.1 (compatible with v2.0) • Refine once a gas cell contains 1*(SPH gas particle mass) or 8 collisionless particles (DM or IC stars) • Experiments were also performed with less aggressive refinement criteria. • The ART-II entry was last updated on Aug. 28, 2016.
CHANGA	Wallace (Quinn)	• Agertz formula with $N_{\text{Jeans}} = 4$ with $\Delta x = 80 \text{ pc}$ (note 1). • In practice, the pressure floor is implemented using the Hopkins et al. 2011 formula (note 2). Our $N_{\text{Jeans}} = 4$ is equivalent to his own $N_{\text{Jeans}} = 8.75$ if h_{sm} is replaced by 80 pc.	• Grackle v2.1 (note 6) • Richie-Thomas SPH forces • Wendland kernel with nSmooth = 64 (note difference from GASOLINE) • Energy Diffusion • Saitoh time stepping • A minimum kernel size to estimate hydrodynamic quantities is set to 0.2 *

		$\text{eps}_{\{\text{grav}\}} = 16 \text{ pc}$. (Here, $\text{eps}_{\{\text{grav}\}}$ is the gravitational softening length = equivalent Plummer softening length = $80 \text{ pc} = h/2.8 = \text{spline size}/2.8$.) i.e. CHANGA parameter $d\text{Soft} = 112 \text{ pc}$, $dh\text{MinOverSoft} = 0.143$ (note 5) • yt-dev or yt-3.2.0 needed for the common yt script to properly work with CHANGA data (PR #1648) • The CHANGA entry was last updated on May 6, 2016.
ENZO	Butler	• Agertz formula with $N_{\text{Jeans}} = 4$ (note 1). i.e. ENZO parameter $\text{MinimumPressureSupportParameter} = 16$ • Grackle v2.0 • PPM reconstruction • HLLC Riemann solver with fallback to HLL solver • Refine once a gas cell contains $1 \times (\text{SPH gas particle mass})$ or $8 \times (\text{IC star particle mass})$ i.e. ENZO parameter $\text{MinimumMassForRefinement} = 0.00008643 \ 0.002765$ with $\text{MassUnits} = 1e9 \text{ Msun}$ • IC input used by Butler is here. • ENZO repository "enzo-dev" changeset cb9712a1f763 is used for this run. • The ENZO entry was last updated on Oct. 11, 2015.
GADGET3	Choi	• Agertz formula with $N_{\text{Jeans}} = 4$ with $\Delta x = 80 \text{ pc}$ (note 1). • In practice, the pressure floor is implemented using the Hopkins et al. 2011 formula (note 2). Our $N_{\text{Jeans}} = 4$ is equivalent to his own $N_{\text{Jeans}} = 8.75$ if $h_{\{\text{sml}\}}$ is replaced by 80 pc. • Grackle v2.1 (note 6) • Quintic kernel with $N_{\text{ngb}}=64$ • A minimum kernel size to estimate hydrodynamic quantities is set to $0.2 \times \text{eps}_{\{\text{grav}\}} = 16 \text{ pc}$. (Here, $\text{eps}_{\{\text{grav}\}}$ is the gravitational softening length = equivalent Plummer softening length = $80 \text{ pc} = h/2.8 = \text{spline size}/2.8$.) i.e. GADGET parameter $\text{Softening}^* = 80 \text{ pc}$, $\text{MinGasHsmlFractional} = 0.0714$ (note 4) • A deviation in the initial temperature is known. We however don't expect that the outcome of the run will be altered much. • Because the code carries only the specific energy field by default, — thus T/μ —, the yt script needs to convert it to actual temperature using the Grackle v2.1 machinery. By the same token, the T/μ field needs to be initially set using the Grackle v2.1 machinery so it corresponds to a constant $T_{\text{init}}=1e4 \text{ K}$ on the disk. To approximately address the first issue, the yt script converts T/μ into actual temperature using a T vs. μ table in Grackle v2.0 (not v2.1, which would have been an ideal solution), as of Mar. 18, 2016. The second issue is still being investigated as of Mar. 2016. The current temperature values in the composite figures above are therefore slightly incorrect. • The GADGET-3 entry was last updated on Sep. 23, 2016.
GASOLINE	Keller (Benincasa)	• Agertz formula with $N_{\text{Jeans}} = 4$ with $\Delta x = 80 \text{ pc}$ (note 1). • Grackle v2.1 (note 6) • Wendland kernel C4 with $n\text{Smooth} = 200$

		• In practice, the pressure floor is implemented using the Hopkins et al. 2011 formula (note 2). Our $N_{\text{Jeans}} = 4$ is equivalent to his own $N_{\text{Jeans}} = 8.75$ if h_{sml} is replaced by 80 pc. i.e. GASOLINE parameter $d_{\text{ResolveJeans}} = 3.06$ (note 3)	(note difference from CHANGA; Dehnen & Aly 2012 showed that this kernel can use larger neighbor numbers without the pairing instability that can effectively removes resolution, and that doing this improves performance on a number of basic hydrodynamics tests) • A minimum kernel size to estimate hydrodynamic quantities is set to $0.2 * \text{eps}_{\text{grav}} = 16 \text{ pc}$. (Here, eps_{grav} is the gravitational softening length = equivalent Plummer softening length = $80 \text{ pc} = h/2.8 = \text{spline size}/2.8$.) i.e. GASOLINE parameter $d_{\text{Soft}} = 112 \text{ pc}$, $dh_{\text{MinOverSoft}} = 0.143$ (note 5) • yt-dev or yt-3.2.0 needed for the common yt script to properly work with GASOLINE data (PR #1648). • The GASOLINE entry was last updated on Jul. 18, 2016.
GEAR	Revaz	• Agertz formula with $N_{\text{Jeans}} = 4$ with $\Delta x = 80 \text{ pc}$ (note 1). • In practice, the pressure floor is implemented using the Hopkins et al. 2011 formula (note 2). Our $N_{\text{Jeans}} = 4$ is equivalent to his own $N_{\text{Jeans}} = 8.75$ if h_{sml} is replaced by 80 pc. i.e. GEAR parameter $\text{JeansMassFactor} = 8.75$	• Grackle v2.0 • Individual and adaptive time steps schemes (Durrier & Dalla Vecchia 2012) • SPH pressure entropy formulation (Hopkins 2013) • Artificial viscosity (Monaghan & Gingold 1983) • Balsara switch (Balsara 1995) • Particle-based time dependent viscosity coefficients (Rosswog et al. 2000) • Original Gadget-2 kernel with $N_{\text{ngb}}=50$ • A minimum kernel size to estimate hydrodynamic quantities is set to $0.2 * \text{eps}_{\text{grav}} = 16 \text{ pc}$. (Here, eps_{grav} is the gravitational softening length = equivalent Plummer softening length = $80 \text{ pc} = h/2.8 = \text{spline size}/2.8$.) i.e. GEAR parameter $\text{Softening}^* = 80 \text{ pc}$, $\text{MinGasHsmfFractional} = 0.0714$ (note 4) • Because the code carries only the internal energy field by default, — thus T/μ —, the yt script converts it to actual temperature using a T vs. μ table in Grackle v2.0. By the same token, the T/μ field needs to be initially set using the T vs. μ table of Grackle v2.0. (e.g. μ is 0.998 at $T = 1\text{e}4\text{K}$, so for the disk, T_{init}/μ should be $1.002\text{e}4$.) These updates were made as of Aug. 17, 2015. • The GEAR entry was last updated on Sep. 25, 2015.
GIZMO	Lupi	• Agertz formula with $N_{\text{Jeans}} = 4$ with $\Delta x = 80 \text{ pc}$ (note 1). • Note that the inter-particle spacing dx in GIZMO is not used. • Note also that the Hopkins et al. 2011 formula (note 2) is not used in GIZMO's implementation of pressure floor.	• Grackle v2.1 (note 6) • Experiments were also performed with other pressure support prefactors, other ways of assigning Δx, and other values for a minimum kernel size for hydro estimation. • A minimum kernel size to estimate hydrodynamic quantities is set to $0.2 * \text{eps}_{\text{grav}} = 16 \text{ pc}$. (Here, eps_{grav} is

		the gravitational softening length = equivalent Plummer softening length = 80 pc = $h/2.8$ = spline size/2.8.) i.e. GIZMO parameter Softening* = 80 pc, MinGasHsm1Fractional = 0.0714 (note 4) - Because the code carries only the internal energy field by default, — thus T/μ —, the yt script needs to convert it to actual temperature using the Grackle v2.1 machinery. By the same token, the T/μ field needs to be initially set using the Grackle v2.1 machinery so it corresponds to a constant $T_{\text{init}}=1e4$ K on the disk. To resolve these two issues, as of Aug. 20, 2015, the GIZMO code has been modified to output the actual temperature from Grackle and to set up a constant temperature, as opposed to internal energy. The corresponding part of the yt script has been modified, too. - The GIZMO entry was last updated on Dec. 14, 2015.
RAMSES	Teyssier (Butler)	- Gas pressure field includes the pressure support by a temperature polytrope. $T_{\text{Jeans}}/\mu = T_J(n_H/n_J)$, where $T_J = 1800$ K, $n_J = 8$ H/cc, and $n_H = \rho * x_H / m_H$ with $x_H = 0.76$ - This polytrope is designed to match the Agertz formula with $N_{\text{Jeans}} \sim 4$. - Grackle v2.0 - PLM reconstruction - RAMSES code carries only the pressure field which includes the Jeans pressure support term. Therefore, a yt script is devised to subtract out the term to retrieve the correct gas internal energy. Note that the exact values of physical constants used in RAMSES's polytrope were used, not the ones in yt's constant library. - Also because the code carries only the pressure field, — thus internal energy or T/μ —, the yt script converts it to actual temperature using a T vs. μ table in Grackle v2.0. By the same token, the T/μ field needs to be initially set using the T vs. μ table of Grackle v2.0. (e.g. μ is 0.998 at $T = 1e4$K, so for the disk, T_{init}/μ should be $1.002e4$.) - There currently is a deviation from the agreed initial temperature of $1e4$K. We however don't expect that the outcome of the run will be altered by much. - RAMSES snapshots have less total number of collisionless particles since its box size, $(320 \text{ kpc})^3$, cuts out particles in the outskirts of potentials. More particles may leave the box and do not contribute. However, the dynamics of a inner disk which we focus on will not be affected. - yt-dev (post yt-3.2.0) needed to properly set RAMSES units (PR #1694). See the PR or the yt issue tracker #1055 for more information. - The RAMSES entry was last updated on Aug. 27, 2015.

[[Note 1](#)] Pressure support formula by Agertz in Dropbox README.txt:

$$P_{\text{jeans}} = 1/\pi * N_{\text{Jeans}}^2 / \gamma * G * \rho^2 * \Delta x^2$$

[Note 2] Pressure support formula by Hopkins et al. 2011 (Eq 1). Note different definition of N_{Jeans} from above:

$$P_{\text{jeans}} = 1.2 * N_{\text{Jeans}}^{2/3} / \gamma * G * \rho^2 * h_{\text{sml}}^2$$

[Note 3] The CHANGA/GASOLINE codes use the Hopkins et al. 2011 formula (Note 2), but their parameter "dResolveJeans" is *not* N_{Jeans} in the Hopkins et al. formula, *but* the whole prefactor $= 1.2 * N_{\text{Jeans}}^{2/3} / \gamma$. Therefore, to set $N_{\text{Jeans}} = 8.75$ in Hopkins et al. formula, dResolveJeans should be set to $= 1.2 * 8.75^{2/3} / (5/3) = 3.06$.

[Note 4] As noted in Section 2. (2–1) above, we define "gravitational softening length" as " eps_{grav} " = "equivalent Plummer softening length" = 80 pc. This equates to the parameters "Softening[Gas/Halo/Disk/Stars]" in GADGET/GEAR/GIZMO. This is different from the typical definition of "spline size h", beyond which the force becomes exactly Newtonian (h as defined in [Eq. 4 of Springel et al. 2005](#)). For GADGET/GEAR/GIZMO, $h = 2.8 * \text{eps}_{\text{grav}} = 224$ pc. Note that the parameter which controls the minimum gas smoothing length, "MinGasHsmfFractional", is defined as a fraction of h, *not* eps_{grav} ($h = \text{ForceSoftening[]}$ in gadget/gravtree.c). Therefore, to make minimum smoothing length $= 0.2 * \text{eps}_{\text{grav}}$, MinGasHsmfFractional should be set to $= 0.2/2.8 = 0.0714$.

[Note 5] In contrast to Note 4, the CHANGA/GASOLINE parameter "dSoft" is *not* the equivalent Plummer softening length, *but* $h/2$ in the Springel et al. 2005 definition of h (see Note 4 above; but keep in mind that dSoft is indeed equal to h defined in some other papers, e.g. [Eq. 14 of Katz et al. 1996](#) or [Eq. 21 of the original Monaghan & Lattanzio 1985](#)). Therefore, to follow the recommendation in Section 2. (2–1), dSoft should be set to $= 224 \text{ pc} / 2 = 112$ pc. Note that the parameter which controls the minimum gas smoothing length, "dhMinOverSoft", is defined as a fraction of dSoft. Therefore, to make minimum smoothing length $= 0.2 * \text{eps}_{\text{grav}}$, dhMinOverSoft should be set to $= 0.2/2.8 * 2 = 0.143$.

[Note 6] Using Grackle v2.1 implies that an updated value for solar metallicity unit $Z_{\text{sun}} = 0.01295$ is used (as opposed to 0.02041 in Grackle v2.0 or below), and that the mean molecular weight is a function of temperature and density (as opposed to a function of temperature only in Grackle v2.0 or below). More discussion [here](#). Initial gas properties – e.g. internal energy, metallicity – should be set up so it is consistent with your chosen Grackle version. Likewise, resulting gas properties – e.g. temperature, metallicity – should be analyzed in a way that is consistent with your Grackle version.

5. "Second" Datasets (Grackle+SF+ThermalFbck)

Before proceeding further, please read the cautionary notes at the beginning of [Section 4](#) above, which also applies here. Make sure to carefully read the Code Notes in both Sections 4 and 5 before drawing any conclusion. As always, the [agora-analysis-script](#) repository is being updated for the analysis presented here (script-Paper4-disk-new.py)

(1) Gas Surface Density Maps

- [0 Myr compilation](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [500 Myr compilation](#) (first posted: 01/11/2016, last update: 09/28/2016)
- Notes: Description of this figure in [Section 4](#)

(2) Temperature Projection Maps

- [0 Myr compilation](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [500 Myr compilation](#) (first posted: 01/11/2016, last update: 09/28/2016)
- Notes: Description of this figure in [Section 4](#)

(3) Gas Metallicity Projection Maps

- [0 Myr compilation](#) (first posted: 03/25/2016, last update: 09/28/2016)
- [500 Myr compilation](#) (first posted: 03/25/2016, last update: 09/28/2016)
- Notes: Density-square weighted. As discussed in [Section 2–\(2\)](#), the disk initial metallicity is set to 0.02041. This corresponds to $1 Z_{\text{sun}}$ in Grackle v2.0 and yt–3.1, but $1.5761 Z_{\text{sun}}$ in Grackle v2.1 and yt–3.2.0+ ([PR #1605](#)). So, to minimize confusion, we present simply the ratio of metal_density/density, not in Z_{sun} . As of Jun. 2016, this map is produced temporarily with a stable yt–3.2.3 (*not* yt–3.3–dev), due to unresolved issues likely related to yt [PR #1740](#). [for 2nd datasets only]

(4) Newly-formed Stellar Particle Maps

- [500 Myr compilation](#) (first posted: 03/28/2016, last update: 09/28/2016)

• Notes: Only the newly-formed stellar particles are drawn -- i.e. *not* the disk or bulge stars in the IC. Raw particle fields are used, not the interpolated or smoothed fields in yt. yt-dev (post yt-3.2.0) needed for the common yt script to properly handle the particle filter and produce this map ([PR #2000](#)). Check "Newly-formed Stellar Particle Map with Clumps" below for stellar clumps analysis. **[for 2nd datasets only]**

[\(5\) Density-Temperature PDF](#)

- [0 Myr compilation](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [500 Myr compilation](#) (first posted: 01/11/2016, last update: 09/28/2016)

• Notes: Description of this figure in [Section 4](#). A thick dashed line is to plot the mean temperature in each density bin for CHANGA's 1st dataset run (Grackle+noSF).

[\(6\) Density-Temperature-Metallicity PDF](#)

- [0 Myr compilation](#) (first posted: 03/25/2016, last update: 09/28/2016)
- [500 Myr compilation](#) (first posted: 03/25/2016, last update: 09/28/2016)

• Notes: Mass-weighted average of gas metallicity in two-dimensional density-temperature bins. Raw particle fields are used for SPH codes, not the interpolated or smoothed fields in yt. As discussed in [Section 2-\(2\)](#), the disk initial metallicity is set to 0.02041. This corresponds to 1 Z_{sun} in Grackle v2.0 and yt-3.1, but 1.5761 Z_{sun} in Grackle v2.1 and yt-3.2.0+ ([PR #1605](#)). So, to minimize confusion, we use the ratio of metal_density/density, not in Z_{sun} . **[for 2nd datasets only]**

[\(7\) Radius-Gas Rotational Velocity PDF + Rotation Curves](#)

- [0 Myr compilation](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [500 Myr compilation](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [500 Myr rotation curves](#) (first posted: 03/25/2016, last update: 09/28/2016)

• Notes: Description of this figure in [Section 4](#)

[\(8\) Radius-Gas Velocity Dispersion](#)

- [500 Myr velocity dispersion curves](#) (first posted: 07/03/2016, last update: 09/28/2016)
- [500 Myr vertical velocity dispersion curves](#) (first posted: 08/22/2016, last update: 09/28/2016)

• Notes: Description of this figure in [Section 4](#)

[\(9\) Radius-New Stellar Particle Rotational Velocity PDF + Rotation Curves](#)

- [500 Myr compilation](#) (first posted: 03/29/2016, last update: 09/28/2016)
- [500 Myr rotation curves](#) (first posted: 03/29/2016, last update: 09/28/2016)

• Notes: Rotational velocity of newly formed stellar particles -- i.e. not the disk and bulge stars in the IC. Radius and velocity with respect to the galactic center (location of maximum gas density within 1 kpc from the center of gas mass for each snapshot). The second figure is simply the composite of all the mass-weighted average lines in the first figure. yt-dev (post yt-3.2.0) needed for the common yt script to properly reproduce this PDF ([PR #1684](#), and [PR #2097](#) for the yt issue tracker [#1203](#)). **[for 2nd datasets only]**

[\(10\) Radius-New Stellar Particle Velocity Dispersion](#)

- [500 Myr velocity dispersion curves](#) (first posted: 07/03/2016, last update: 09/28/2016)
- [500 Myr vertical velocity dispersion curves](#) (first posted: 08/22/2016, last update: 09/28/2016)

• Notes: Velocity dispersion of newly formed stellar particles -- i.e. not the disk and bulge stars in the IC. This quantifies the residual velocity components other than the rotational velocity calculated above; in other words, square roots of mass-weighted averages of $[v_i - v_{\text{rot, local}}]^2$. Radius and velocity with respect to the galactic center (location of maximum gas density within 1 kpc from the center of gas mass for each snapshot). The second figure shows only the vertical velocity dispersion to illustrate the contribution in dispersion by the vertical stellar movement. yt-dev (post yt-3.2.0) needed for the common yt script to properly produce this PDF ([PR #1684](#), and [PR #2097](#) for the yt issue tracker [#1203](#)). **[for 2nd datasets only]**

[\(11\) Radius–Gas Vertical Height PDF + Radially–binned Average Gas Vertical Height](#)

- [0 Myr compilation](#) (first posted: 07/01/2016, last update: 09/28/2016)
- [500 Myr compilation](#) (first posted: 07/01/2016, last update: 09/28/2016)
- [500 Myr average gas height curves](#) (first posted: 07/01/2016, last update: 09/28/2016)
- Notes: Description of this figure in [Section 4](#)

[\(12\) Gas Mass Distribution in Density \(Density PDF\)](#)

- [0 Myr compilation](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [500 Myr compilation](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [500 Myr difference between 1st and 2nd datasets](#) (first posted: 08/22/2016, last update: 09/28/2016)
- Notes: Description of this figure in [Section 4](#). The third figure shows the change in mass in each density bin from the density PDF of the 1st dataset (see Section 4 above) to the that of the 2nd dataset (second figure right above). This measures the effect and star formation and feedback. To make smaller changes to stand out, the bottom panel of the third figure presents $\text{symlog}[\text{mass change}]$, where $\text{symlog}(x) = \text{sign}(x)\log|x|$.

[\(13\) Gas Mass Distribution in Radius + Radially–binned Gas Surface Density](#)

- [0 Myr compilation – mass](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [500 Myr compilation – mass](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [0 Myr compilation – surface density](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [500 Myr compilation – surface density](#) (first posted: 01/11/2016, last update: 09/28/2016)
- Notes: Description of this figure in [Section 4](#)

[\(14\) New Stellar Mass Distribution in Radius + Radially–binned New Stellar Surface Density](#)

- [500 Myr compilation – mass](#) (first posted: 03/29/2016, last update: 09/28/2016)
- [500 Myr compilation – surface density](#) (first posted: 03/29/2016, last update: 09/28/2016)
- Notes: The first figure shows newly formed stellar mass distribution (accumulated) along the cylindrical radius axis. The second figure is the stellar surface density profile as a function of cylindrical radius, simply derived from the first figure. Radius with respect to the galactic center (location of maximum gas density within 1 kpc from the center of gas mass for each snapshot). **[for 2nd datasets only]**

[\(15\) Radially–binned Star Formation Rate Surface Density](#)

- [500 Myr compilation – SFR surface density](#) (first posted: 07/04/2016, last update: 09/28/2016)
- Notes: Star formation rate surface density as a function of cylindrical radius. Star formation rate is estimated using star particles that are younger than 20 Myrs old. Radius with respect to the galactic center (location of maximum gas density within 1 kpc from the center of gas mass for each snapshot). **[for 2nd datasets only]**

[\(16\) Kennicutt–Schmidt Plot – Using Azimuthally–Averaged Profiles](#)

- [500 Myr Kennicutt–Schmidt plot](#) (first posted: 07/04/2016, last update: 09/28/2016)
- [500 Myr Kennicutt–Schmidt plot with fits](#) (first posted: 07/04/2016, last update: 09/28/2016)
- Notes: Kennicutt–Schmidt plot using azimuthally–averaged gas and star formation rate (SFR) surface densities found above. Check "Gas Surface Density" and "Star Formation Rate Surface Density" above to see how each quantities were calculated. Only the annuli with nonzero averaged SFR surface densities are considered. The thick black dashed line denotes a best observational fit by Kennicutt et al. (2007). The blue contour marks the distribution of sub–kpc patches in nearby galaxies found in Figure 8 of Bigiel et al. (2008), where their hydrogen surface density is multiplied by 1.36 to account for helium and match the total gas density in our simulations. The second figure is simply the same plot as the first one with linear fits for each codes. **[for 2nd datasets only]**

[\(17\) Kennicutt–Schmidt Plot – Using Sub–kpc Patches](#)

- [500 Myr gas surface density map at 750 pc resolution](#) (first posted: 08/22/2016, last update: 09/28/2016)

- [500 Myr SFR surface density map at 750 pc resolution](#) (first posted: 08/22/2016, last update: 09/28/2016)
- [500 Myr Kennicutt–Schmidt plot – using patches](#) (first posted: 08/22/2016, last update: 09/28/2016)

• Notes: The first figure is the face-on gas surface density at 750 pc resolution. For particle-based codes, surface densities are estimated by depositing gas particles by CIC onto a 2-dimensional uniform grid with 750 pc resolution. This image could be considered as a degraded version of “Gas Surface Density Maps” above although a different deposit algorithm is used. The second figure is the face-on SFR surface density at 750 pc resolution. SFR surface densities are estimated by depositing the newly created star particles that are younger than 20 Myrs old onto the same uniform grid. The third figure is the Kennicutt–Schmidt plane, same as (16), but now with gas surface densities and SFR surface densities averaged in each 750 pc × 750 pc patches. As an example, shown as gray triangles are the patches with non-zero SFR surface density for CHANGA. An 80% percentile contour is drawn for each code for all such patches. The thick black dashed line denotes a best observational fit by Kennicutt et al. (2007). The blue contour marks the distribution of sub-kpc patches in nearby galaxies found in Figure 8 of Bigiel et al. (2008), where their hydrogen surface density is multiplied by 1.36 to account for helium and match the total gas density in our simulations. **[for 2nd datasets only]**

[\(18\) Gas Mass Distribution in Height + Vertically-binned Gas Surface Density](#)

- [0 Myr compilation – mass](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [500 Myr compilation – mass](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [0 Myr compilation – surface density](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [500 Myr compilation – surface density](#) (first posted: 01/11/2016, last update: 09/28/2016)

• Notes: Description of this figure in [Section 4](#)

[\(19\) Gas Cell Size Maps](#)

- [0 Myr compilation](#) (first posted: 01/11/2016, last update: 09/28/2016)
- [500 Myr compilation](#) (first posted: 01/11/2016, last update: 09/28/2016)

• Notes: Description of this figure in [Section 4](#)

[\(20\) Gas Resolution Maps](#)

- [0 Myr compilation](#) (first posted: 07/01/2016, last update: 09/28/2016)
- [500 Myr compilation](#) (first posted: 07/01/2016, last update: 09/28/2016)

• Notes: Description of this figure in [Section 4](#)

[\(21\) Gas Disk Elevation Maps](#)

- [0 Myr compilation](#) (first posted: 06/30/2016, last update: 09/28/2016)
- [500 Myr compilation](#) (first posted: 06/30/2016, last update: 09/28/2016)

• Notes: Description of this figure in [Section 4](#)

[\(22\) Newly-formed Stellar Particle Maps with Clumps Annotated + Stellar Clumps Mass Spectra](#)

- [500 Myr stellar particle map with clumps annotated – HOP](#) (first posted: 07/08/2016, last update: 09/28/2016)
- [500 Myr stellar particle map with clumps annotated – FOF](#) (first posted: 07/08/2016, last update: 09/28/2016)
- [500 Myr stellar clump mass spectrum – HOP](#) (first posted: 07/08/2016, last update: 09/28/2016)
- [500 Myr stellar clump mass spectrum – HOP/cumulative](#) (first posted: 07/11/2016, last update: 09/28/2016)
- [500 Myr stellar clump mass spectrum – FOF](#) (first posted: 07/08/2016, last update: 09/28/2016)
- [500 Myr stellar clump mass spectrum – FOF/cumulative](#) (first posted: 07/11/2016, last update: 09/28/2016)

• Notes: First two figures are the same stellar map shown in (4), but with annotated stellar clumps identified by HOP or FOF. The next plots reveal clump mass functions. Clumps with masses above $2.6 \times 10^6 M_{\text{sun}}$ ($\sim 30 \times$ minimum stellar particle mass at formation = $30 \times 8.593 \times 10^4 M_{\text{sun}}$) are shown. Only the newly-formed stellar particles are considered when identifying clumps -- i.e. *not* the disk or bulge stars in the IC. The cumulative mass functions include an additional test GIZMO–PS2 for which GIZMO’s pressure floor is increased by a factor of 2. yt-dev (post yt–3.2.0) needed to enable HOP/FOF to take stellar particles as input ([PR #2085](#)). A patch is applied in yt to make HOP/FOF work for non-cosmological data. See the [agora-analysis-script](#) for details. **[for 2nd datasets only]**

[\(23\) Star Formation Rates](#)

- [Compilation by 500 Myr – SFR](#) (first posted: 05/04/2016, last update: 09/28/2016)
- [Compilation by 500 Myr – accumulated total stellar mass](#) (first posted: 05/04/2016, last update: 09/28/2016)

• Notes: Using the creation time field tagged for each stellar particles in 500 Myr datasets. The second figure is simply the time integrated version of the first figure. 25 bins along the time axis (x-axis). The SFR plot includes an additional test GIZMO-PS2 for which GIZMO's pressure floor is increased by a factor of 2. yt-dev (post yt-3.2.0) needed for the common yt script to properly handle the particle creation time fields, and reproduce these figures (**PR # to be added**). In particular, see the Code Notes for ART-I below. **[for 2nd datasets only]**

[\(24\) More Diagnostics to Come](#)

•

[\(25\) Code Notes](#)

CODE	SUBMITTED BY	NOTES
		(Only the items unique to SF+fbck runs are included below, so readers are encouraged to check the Code Notes in Section 4 for general descriptions. Recent updates in blue.)
ART-I	Ceverino	• Efforts were made so that the thermal feedback scheme is properly handled in ART-I when using Grackle. • As noted in the star formation rates figures, as of May 2015, yt-3.3-dev does not properly handle ART-I's particle_creation_time field (or tbirth field) in a non-cosmological simulation, so a fix needs to be applied in yt. See the agora-analysis-script for details. A proper pull request will be issued soon, and posted in this page. • Special care needs to be taken to properly interpret the metallicity in the ART-I data. See the agora-analysis-script for details. • The ART-I entry was last updated on Sep. 28, 2016.
ART-II	Feldmann	• Following experiments have also been performed with less aggressive refinement criteria: (i) SF+noFbck, (ii) SF+ThermalFbck (standard), (iii) SF+ThermalFbck+delayed cooling for 40 Myr. • The ART-II entry was last updated on Aug. 28, 2016.
CHANGA	Wallace (Quinn)	• AGORA feedback scheme needs to be implemented differently than is normally done in CHANGA in which normally an energy injection rate is set. The AGORA feedback scheme is instead done by instantaneously changing the thermal energy of the surrounding gas. • Attention needs to be paid to the CHANGA star formation prescription to exactly follow the AGORA recommendation. i.e. CHANGA parameter CStar = 0.0052 (note 1) • Hot bubbles in the temperature projection maps (or a group of hot, dense gas in the density-temperature phase plot) has been known. It was found that smoothing over a larger number of particles (i.e. increasing nSmooth from 64 to 200 that matches GASOLINE's) gets rid of this feature. • A time stepping fix was developed and applied to accommodate relatively swift production of supernova feedback energy. • The CHANGA entry was last updated on May 6, 2016.
ENZO	Butler (Kim)	• AGORA star formation and feedback scheme needs to be implemented differently than is normally done in any of ENZO's star_maker#.F routines. In this particular case, star_maker4.F was edited to model the Schmidt-like star formation, and instantaneous thermal feedback with 5 Myr time delay. • ENZO parameter file for this run is here. • The ENZO entry was last updated on May 30, 2016.
GADGET3	Choi	• Because the code carries only the specific energy field by default, — thus T/mu —, the yt script needs to convert it to actual temperature using the Grackle v2.1 machinery. By the same token, the T/mu field needs to be initially set using the Grackle v2.1 machinery so it corresponds to a constant T_init=1e4 K on the disk. To approximately address the first issue, the yt script converts T/mu into actual temperature using a T vs. mu table in Grackle v2.0 (not v2.1, which would have been an ideal solution). The second issue is still being investigated as of Mar. 2016. The current temperature values in the composite figures above are therefore slightly incorrect. • A deviation in the initial temperature is known. We however don't expect that the

		outcome of the run will be altered much. • A distinctive method developed by K. Todoroki is used to couple SN thermal feedback energy with its surroundings model: first, estimate the SN shock radius based on Sedov–Taylor solution from the location of young stars, then deposit the thermal energy to the gas particles within the shock radius. This radius depends on energy and gas density but with weak powers. In a pure Sedov–Taylor model only a small number of gas particles are found in this shock radius (20–50 pc), whereas in a revised model as in Stinson et al. 2006 or Chevalier 1974 -- adopted in the current entry -- a larger number of particles are found. Other variations such as Draine have also been tested. • This feedback scheme may sometimes result in a small number of gas particles moving at ~ 1000 km/s. In the actual cosmological simulations, these wind particles would interact with the IGM and slow down to few 100km/s, but in our current sim, the resolution is not so high and the wind is expanding into almost vacuum unabated. • This feedback scheme also assigns some parts of returning gas mass to star particles, which may lead to stellar particle masses deviating slightly from a fixed value. We however don't expect that the outcome of the comparison will be affected much. • Experiments were also carried out with (1) thermal+kinetic feedback, and (2) a multiphase ISM model (Springel & Hernquist 2003) to describe thermal feedback. • The GADGET-3 entry was last updated on Sep. 23, 2016.
GASOLINE	Keller	• AGORA feedback scheme needs to be implemented differently than is normally done in GASOLINE in which normally an energy injection rate is set. The AGORA feedback scheme is instead done by instantaneously changing the thermal energy of the surrounding gas (using the smoothing kernel; see Code Notes in Section 4 above). • Attention needs to be paid to the GASOLINE star formation prescription to exactly follow the AGORA recommendation. i.e. GASOLINE parameter CStar = 0.0052 (note 1) • The typical centering scheme (location of maximum gas density within 1 kpc from the center of gas mass for each snapshot) does not find the visual galactic center for the current 500 Myr entry. We temporarily define the center of the most massive stellar clump as the galactic center for this particular snapshot. • The GASOLINE entry was last updated on Jul. 20, 2016.
GEAR	Revaz	• Energy and metal deposition is performed among the 50 nearest neighbours, weighted according to the SPH kernel. • Because the code carries only the internal energy field by default, — thus T/μ —, the yt script converts it to actual temperature using a T vs. μ table in Grackle v2.0. By the same token, the T/μ field needs to be initially set using the T vs. μ table of Grackle v2.0. (e.g. μ is 0.998 at $T = 1e4K$, so for the disk, T_{init}/μ should be $1.002e4$.) • An experiment was also performed with a delayed cooling (adiabatic time of 5 Myr) to describe SN thermal feedback, along with SN rates computed from a Kroupa IMF (Kroupa 2001; not Chabrier). • Special care needs to be taken to properly read the metal field in the GEAR format. See the agora-analysis-script for details. • The GEAR entry was last updated on Jul. 4, 2016.
GIZMO	Lupi	• Because the code carries only the internal energy field by default, — thus T/μ —, the yt script needs to convert it to actual temperature using the Grackle v2.1 machinery. By the same token, the T/μ field needs to be initially set using the Grackle v2.1 machinery so it corresponds to a constant $T_{init}=1e4$ K on the disk. To resolve this, the GIZMO code has been modified to output the actual temperature from Grackle and to set up a constant temperature, as opposed to internal energy. • Experiments were also performed with other feedback schemes and other values of disk initial metallicity. • The GIZMO entry was last updated on Mar. 23, 2016.
RAMSES	Teyssier	• Because the code carries only the pressure field, — thus internal energy or T/μ —, the yt script converts it to actual temperature using a T vs. μ table in Grackle v2.0. By the same token, the T/μ field needs to be initially set using the T vs. μ table of Grackle v2.0. (e.g. μ is 0.998 at $T = 1e4K$, so for the disk, T_{init}/μ should be $1.002e4$.) • There currently is a deviation from the agreed initial temperature of $1e4K$. We however don't expect that the outcome of the run will be altered by much

- Note that particle_age field actually means particle creation time, at least for this particular dataset. See the [agora-analysis-script](#) for details.
- Experiments were also performed with other feedback channels such as radiation.
- The RAMSES entry was last updated on Jun. 18, 2016.

[Note 1] Note that the CHANGA/GASOLINE parameter "CStar" (defined in [Eq. 3 of Katz 1992](#)) is *not* equal to e_{ff} defined in Section 2. (2-2) above (or [Eq. 2 of the Flagship Paper](#)). This is due to the difference in the denominator of the Schmidt formula: local gas dynamical time in Katz 1992, $t_g = \sqrt{1/(4\pi G \rho_{\text{gas}})}$, versus local free-fall time in the Flagship Paper, $t_{\text{ff}} = \sqrt{3\pi/(32 G \rho_{\text{gas}})}$. Therefore, to follow the recommendation in Section 2. (2-2), CStar should be set to $e_{\text{ff}} * (t_g / t_{\text{ff}}) = 0.01 * \sqrt{32/12/\pi^2} = 0.0052$.

6. Finalizing Paper "4" (In the 5th Workshop) + All the Links You Need

(1) 5th AGORA Workshop

- Workshop program (Aug. 12-14, 2016) : [LINK](#)
- Collaboration-wide web conference agenda (Aug. 14, 2016) : [LINK](#)

(2) Paper 4 Draft

- Draft that has been reviewed by the core writer group and code leaders : [LINK](#)
- The current version above will be kept in place until Aug. 21 so that any revision wouldn't cause confusion to section writers. See (3) below. Note that this draft will significantly change after Aug. 21 with reorganized and revamped sections, new figures, new styles, etc. (OLD INFORMATION)
- Status of each Sections (OLD INFORMATION)
 - *Section 1. Introduction* : Currently contains only the obvious factual statements that we think needs be there, not the motivation of this work. Some of us including Joel are working on it, but certainly welcome your input.
 - *Sections 2-5* : These 4 sections are close to be final. Here we assembled sufficient details to reproduce our comparison — IC, common physics, common runtime parameters, codes, etc — hoping to make our test a gold standard for future codes.
 - *Section 6. Results* : Currently contains mostly the factual statements about how each analyses are carried out. Before and during the Workshop we will think about how to tell the story in the most efficient and organized fashion, jointly discuss the diagnostics, and pick and write unfinished subsections on the spot.
- We kindly remind you of the [internal policies](#) regarding the disclosure of AGORA simulation results, and regarding the authorships of AGORA publications. Although this page is accessible by any AGORA members, at this point this draft is *not* being advertised outside the AGORA Paper Group 4.
- Datasets used in the draft
 - All entries are final at this point barring unexpected changes; as of Aug. 2016
 - All figures and diagnostics for 1st datasets (Grackle+noSF) : See [Section 4](#) above
 - All figures and diagnostics for 2nd datasets (Grackle+SF+ThermalFbck) : See [Section 5](#) above

(3) Writing Unfinished Subsections of "Section 6. Results"

- Learn what was discussed in the Workshop by going to this [Google Doc](#). It transparently describes all the topics discussed during the 5th Workshop.
- Paper Group 4 participants are highly encouraged to sign up and write unfinished subsections that interests you the most. Look for pink colored texts to sign up for each subsections and leave your name there so we can track who contributed to which parts. Use the figure-by-figure discussion points we have collected. Ideally, you may want to turn on "Suggesting" instead of "Editing" to clearly mark your contributions. The

Section 6's text will be open to your contribution **until Aug. 21, 2016**.

[7. Planning of the AGORA Paper "4b" or "Disk-II"](#)

- To be elaborated here in the very near future!
- In the meantime, please see the [discussion note](#) during our 5th Workshop (Aug. 2016).